

Bhalaji Yadav Kantepalle

Richmond, VA | | contact@bhalaji.com
linkedin.com/in/kbhalajiyadav | Portfolio: bhalaji.com

PROFESSIONAL SUMMARY

Chemical Engineering Professional (M.S. Candidate) specializing in **Quantitative Metrology**, **Soft Matter Mechanics**, and **Process Automation**. Expertise lies in bridging the gap between R&D and Manufacturing—using tools like **Python (OpenCV)** and **Rheology** to solve complex material problems, while leveraging **SAP ERP** and **FDA Compliance** frameworks to ensure scalability. Proven track record of securing research funding (\$30k), compressing operational timelines by >90%, and driving digital transformation in regulated environments.

EDUCATION

Virginia Commonwealth University (VCU)

M.S., Chemical & Life Science Engineering | GPA: 3.62

Thesis: "Beyond Peel Strength: Stability-Driven Adhesion Metrics for Soft Wearable Interfaces"

Focus: Fracture Mechanics, Mechanochromic Sensors, Process Optimization, Experimental Design (DOE).

Achievement: Graduate Assistantship Award (Fall 2025) – Full Tuition Waiver & Stipend.

Richmond, VA

Exp. Dec 2026

Amrita University

B.Tech., Chemical Engineering

Coimbatore, India

Jun 2022

TECHNICAL SKILLS

Materials Characterization: Fracture Mechanics, Adhesion Testing (ASTM D2724), Rheology (Cross/Power-law), Optical Microscopy

Computational R&D: Python (OpenCV, Pandas, SciPy, NumPy), Docker, Image Processing, CIE L*a*b* Colorimetry

Operational Excellence: SAP ERP Validation (OQ/PQ), V-Model Framework, Lean Manufacturing, RCA, CAPA

Regulatory Compliance: FDA 21 CFR Part 11, cGMP (ICH Q7 / US FDA), FDA Class I Mapping, GDP

RESEARCH EXPERIENCE

VCU Soft Functional Materials Lab

Graduate Researcher & Metrology Lead

Richmond, VA

Sep 2024 – Present

- Novel Optical Metrology:** Engineered a high-throughput **Computer Vision pipeline** (Python/OpenCV) to quantify mechanochromic response in smart textiles. Synchronized sub-pixel texture tracking with **CIE L*a*b* colorimetry**, increasing metrology throughput by **300%**.
- Fracture Mechanics Automation:** Developed a first-principles analysis protocol to identify "false positive" adhesion failures. Automated the extraction of **Peel Stability Index (PSI)** metrics, reducing data processing time by **>80%**.
- Material Characterization:** Characterized viscoelastic properties of cholesteric LCs using **Cross and Power-law rheology models**. Correlated shear history to helical pitch using **Abbe Refractometry** to optimize optical sensitivity.
- Grant Success (\$30,000):** **Drafted** a successful research proposal for the Commonwealth Cyber Initiative (CCI CVN). **Substantiated** the clinical viability of thermochromic textiles through **market gap analysis** and **FDA Class I regulatory mapping**.

INDUSTRY EXPERIENCE

Kreative Organics Pvt. Ltd.

Technical Project Manager

Hyderabad, India

May 2023 – May 2024

- SAP System Requalification:** Re-engineered the deployment strategy to meet a strict 24-hour executive deadline. Compressed the standard 5-day timeline into a **5-hour overnight operation (>90% reduction)**, achieving zero production downtime.
- FDA Compliance & Quality Systems:** Led the **Quality Assurance interface** for SAP ERP implementation. Translated manufacturing SOPs into digital validation protocols (**OQ/PQ**), ensuring system logic enforced **cGMP** safety controls.

Technical Project Intern

Oct 2022 – Apr 2023

- Chemical Data Strategy:** **Spearheaded** the initiative to standardize CAS/IUPAC nomenclature across global trade databases. **Directed** the "Chemical Logic" algorithm and **engineered** Docker/Tableau workflows, reducing manual research time by **70%**.

PUBLICATIONS & ACHIEVEMENTS

Journal Article: I. Caloian, J. Trapp, **B. Y. Kantepalle**, et al. "Mechanical Properties of Dual-Layer Electrospun Fiber Mats." *Polymers* 17(13), 1777 (2025). doi:10.3390/polym17131777

Selected Participant: 1st National Neutron Scattering School, Oak Ridge National Laboratory (Sep 2025).

Poster Presentation: "Adhesives for Personalized Wearable Devices" (ACS Fall 2025, Washington D.C.).